TOPOLOGICALLY NILPOTENT ELEMENTS FORM A CLOSED IDEAL, AND THEIR BEHAVIOUR UNDER CONTINUOUS MAPS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof contains the author's sentence 'We omit the proof of the fact that b is a radical ideal (good exercise in the definitions).' It is reproduced verbatim and recorded as an obligation of the card: the omitted item is a named, specific verification, while closedness of the set and both numbered assertions are proved in the span.

Editorial note. The counted spans carry 1 navigation stub; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The closure of a subset of a linearly topologized ring is the set of elements congruent to elements of the subset modulo every open ideal. Same chapter, Lemma lemma-closed, invoked at proof lines 1152-1154. Identifies an element of the closure as a sum of a topologically nilpotent element and an element of any given open ideal.
- Topological nilpotence of an element of a linearly topologized ring. Standard definition, in force in the statement. Fixes the property investigated and the form of the verification.

Source reference. Original lemma 0AMS: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 922–922. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let B be a linearly topologized ring. The set of topologically nilpotent elements of B is a closed, radical ideal of B. Let $\varphi : A \to B$ be a continuous map of linearly topologized rings.*

(1) *If $f \in A$ is topologically nilpotent, then $\varphi(f)$ is topologically nilpotent.*
(2) *If $I \subset A$ consists of topologically nilpotent elements, then the closure of $\varphi(I)B$ consists of topologically nilpotent elements.*

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Proof. Let $\mathfrak{b} \subset B$ be the set of topologically nilpotent elements. We omit the proof of the fact that $\mathfrak{b}$ is a radical ideal (good exercise in the definitions). Let g be an element of the closure of $\mathfrak{b}$. Our goal is to show that g is topologically nilpotent. Let $J \subset B$ be an open ideal. We have to show $g^e \in J$ for some $e \geq 1$. We have $g \in \mathfrak{b} + J$ by Lemma 0AMS. Hence $g = f + h$ for some $f \in \mathfrak{b}$ and $h \in J$. Pick $m \geq 1$ such that $f^m \in J$. Then $g^{m+1} \in J$ as desired.

Let $\varphi : A \to B$ be as in the statement of the lemma. Assertion (1) is clear and assertion (2) follows from this and the fact that $\mathfrak{b}$ is a closed ideal. $\square$